

Intake Scale Plan

How the online data path grows from today's tiny demo to a real cohort, and later to production traffic. It includes proposed answers to your PR #17 questions, the numbers behind them, and the next steps if volume grows. Everything here is yours to push back on.

WHAT I NEED FROM YOU

- 1. **Decide the Gold cohort and marts** (section 2). My proposals are starting points, not decisions.
- 2. **Confirm the Bronze-side findings you own** in `DATA_QUALITY.md`: DF-002, DF-004, DF-014, DF-015. Also write the freshness (update) policy, which the brief requires.
- 3. **Review ADR-004's capacity figures**, which need the two corrections in section 3, **and ADR-005/006 as a decider**. Then approve [PR #21](#).

Please stay unexposed to the test set. You're the only one on the team who hasn't seen a frozen test case, so you approve any behaviour change to the extractor (ADR-006, decision 5). In `evals/intake/frozen_es_pt_v1/`, read only `POLICY.md` and the current `REVIEW_STATUS.md`, and don't open older commits of that file.

1 · WHERE WE ARE

Today's numbers

MEASURE	VALUE	SOURCE
Resources per customer episode	7 req · 13 queries · 165 read · 10 written	ADR-004, measured on local D1
Worker CPU per request	0–4 ms	Production, 29 Sep
D1 round trip (Worker in GRU, D1 in eastern US)	~148 ms median	Production, 29 Sep
Customer path server time	~1.4 s	Production, one episode
Free-plan ceiling (rows written binds first)	10,000 episodes/day	ADR-004
Observed demand in the dataset	17–818 episodes/day	ADR-004 scenarios S1–S3

MEASURE	VALUE	SOURCE
Hosting cost	\$0 (Paid would be \$5/month)	ADR-004
One transaction row on the D1 schema, with indexes	236 bytes	Measured today on the real migrations
Production D1 contents	3 customers · 4 tx · 2 cases · 94 KB	Remote D1, 29 Sep

2 · YOUR PR #17 QUESTIONS

Proposed answers

QUESTION	PROPOSAL
Cohort	1,000 customers stratified by country (MX 50 / CO 30 / AR 20) and segment (60/25/10/5), each with at least one approved purchase. About half have an <code>Cargo no reconocido</code> complaint. That link works at customer level only, because complaints never point to the right product (DF-002) or amount (DF-003).
Gold marts and grain	One row per approved card purchase, which is V1's scope. One profile per customer: country, segment, cards with last four digits. The context card. Contact and digital history become per-customer aggregates, not event rows.
Boundaries	Online and per customer: purchases, profile, card. Precomputed: bank-wide context such as demand and rates. "One Gold module → one API module → one UI component per domain" works, as long as all SQL stays in <code>back-end/src/store/d1.js</code> (ADR-003).
Reconciliation and freshness	Cohort counts reconcile to Silver by business timestamp, never <code>process_date</code> : DF-004 shows a 6–8 hour offset. Seeds are already content-versioned. Rebuild when a source partition changes, on a written schedule (daily is enough for the demo).
Capacity	ADR-004 holds, with the two corrections below.

3 · CAPACITY

What fits, and two corrections to ADR-004

SERVED DATA	ROWS	D1 SIZE	WRITES TO LOAD ($\approx 3/\text{ROW}$)	FREE PLAN
Today	4	94 KB total	—	fits
1,000 customers, approved purchases	$\sim 6.6\text{k}$	~ 1.6 MB	$\sim 20\text{k}$	fits in a day
1,000 customers, all transactions	$\sim 29.5\text{k}$	~ 7 MB	$\sim 90\text{k}$	just fits
All 150k customers, approved purchases	$\sim 996\text{k}$	~ 235 MB	$\sim 3\text{M}$	size fits, daily writes don't
All 4.4M transactions	4.4M	~ 1 GB	$\sim 13\text{M}$	needs Paid (10 GB)

- **The initial load hits the daily write quota, not the size cap.** Free allows 100,000 rows written a day. Anything beyond a $\sim 1,000$ -customer cohort needs Workers Paid, or a load spread over several days.
- **Runtime events will roughly double writes per episode** ($\sim 10 \rightarrow \sim 20$), which lowers the Free ceiling to about 5,000 episodes/day. ADR-004 should say so.

4 · RECOMMENDATION

The simplest approach that works

Keep the current stack and add no infrastructure:

- Worker + D1 on the Free plan, with a **1,000-customer Gold cohort** (2–7 MB, loaded in one day).
- No cache. Idempotency stays in the database (`UNIQUE` + `ON CONFLICT`), and the reference is issued only after the case is read back.
- Runtime events in an append-only D1 table.
- The model on Workers AI, within the free allocation, for the demo and the evaluation.

All of it fits with room to spare, at \$0.

5 · LIMITS WE STATE OPENLY

Limitations

- **One write primary, outside Latin America.** Distance dominates latency, at about 150 ms per round trip. D1 location hints offer no South American region, which is a data-residency problem for a real bank.
- **Free-plan limits:** 500 MB, 100k writes a day, 3-day log retention, and few model calls a day.
- **No load test.** The dataset's demand is synthetic with a flat hourly profile, so peaks are assumptions.
- **Demo identity only.** The team gate and simulated login aren't customer authentication.

- **A second report of the same purchase under a new key isn't blocked yet.** ADR-004 already lists this cross-key duplicate rule as remaining work.

6 · IF TRAFFIC GROWS

Next steps by trigger

TRIGGER	ACTION
>70% of any Free daily limit for 3 days, or >5,000 episodes/day with events	Workers Paid, \$5/month: 10 GB D1 and 50M writes/month.
Read latency	Confirm Smart Placement, and use D1 read replicas (Sessions API) for lists and the agent view. Writes stay on the primary, so case creation keeps the same consistency.
D1 over 5 GB, or write p95 over 200 ms	Split D1 by country (D1 is built for many small databases), or move to PostgreSQL through Hyperdrive, ADR-003's exit.
Event volume	Metrics go to Analytics Engine, or to Queues → R2 as Parquet. D1 keeps only operational rows.
Duplicate reports	A partial unique index on (customer, transaction) where the case is open, which returns the existing reference.
Real production	Bank login (OIDC), WAF rate limiting, data residency, an immutable audit log, retention and data-subject requests, a load test with SLOs, and data-handling approval for the model provider.
Pipeline	Incremental Gold per partition, with automatic reconciliation to Silver and the written freshness policy.

Right principle, wrong mechanism

Keep

- **Consistency over availability for case creation.** We already behave that way: one D1 primary, fail closed with "retry with the same key", the reference only after read-back, and 10 parallel retries producing one case.
- **A short consistency section in ADR-003.** Judges value explicit trade-offs.
- **The one-open-case-per-purchase rule.** This is the real gap the idea points at.

Drop

- **Redis as the idempotency lock.** It becomes a second source of truth, a retry after the TTL expires duplicates anyway, and a crash can leave the key stuck in "processing". Workers KV is eventually consistent, so it can't be a lock either.
- **customer + transaction as the retry key.** It mixes a retried request with a new report about the same charge.
- **Caching customer and card data.** The online path never reads Bronze or Silver, and the cache would spread PII for no gain.

Sources: ADR-003, ADR-004, ADR-006, [DATA_QUALITY.md](#) (design-window figures), production D1 and Workers logs on 29 Sep 2026, and a size measurement on the real D1 migrations. Row counts for the cohort are estimates from Silver averages: about 6.6 approved purchases and 29.5 transactions per customer over three years.